

Automated Regression Engine & Modeling UI for CPG Volume Sales

"When one model just isn't enough: application of automated modeling techniques to high-granularity datasets with UI integration."

Week 2 Project Deliverables

Business Case · Project Proposal · Detailed Requirements · Testing Plan

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1 Business Case

Big Chalk Analytics builds marketing-mix and sales-driver models for consumer packaged goods (CPG) clients. This project delivers an **automated regression engine** that fits a defensible Volume Sales model for every Brand $\times$ Retailer-Channel combination in a client's portfolio, wrapped in an interactive UI that lets analysts review, constrain, and finalize each model without writing code.

1.1 The problem & opportunity

- **Modeling does not scale to the client's granularity.** The client is a national CPG manufacturer with many niche and flavored products sold across up to nine retailer channels. Volume Sales behaves differently for each Brand $\times$ Channel slice, so a single pooled model is inadequate — yet hand-building hundreds of bespoke regressions per refresh cycle is infeasible for a lean analytics team.
- **Manual modeling is slow, inconsistent, and hard to audit.** When each model is crafted by hand, variable selection, sign logic, and diagnostics vary from analyst to analyst. This makes results difficult to reproduce, compare across slices, or defend to the client.
- **Automation alone is not trusted.** Fully automated, black-box selection (e.g., pure regularization) routinely produces coefficients with the *wrong business sign* — negative media effects, positive price effects — which destroys client confidence and cannot be shipped.
- **The opportunity:** a repeatable engine that combines automated selection with *business-sign discipline* and a human-in-the-loop UI lets Big Chalk model an entire portfolio in a fraction of the time, with consistent diagnostics and full auditability — turning a multi-week manual effort into a same-day, reviewable workflow.

1.2 Current situation

- Big Chalk receives client data as weekly point-of-sale and marketing files at the Brand $\times$ Channel $\times$ Week grain (≈ 3 years of history, up to ~ 113 columns per brand spanning price, distribution, trade, media, macro, and competitive families).
- Models are presently built one slice at a time in analyst scripts; there is no standardized, reusable pipeline that guarantees leakage control, adstock treatment, and sign-constrained final fits across all slices.
- There is no self-serve interface: a non-technical reviewer cannot rerun a model, toggle a variable, or tighten a coefficient bound without an engineer in the loop.
- Deliverables the client expects — coefficients, fit metrics, variable contributions ("due-tos"), VIF, and t-stats — are assembled manually per model.

1.3 Constraints & risks to success

- **Confidentiality (NDA).** The dataset is anonymized and covered by a Big Chalk NDA; it must not be redistributed and is excluded from any public code repository.

- **Methodological constraint.** The *final* model must support per-coefficient constraints (positive / negative / unconstrained / a custom range that excludes zero). This rules out a pure Ridge estimator for the final fit, since Ridge cannot honor sign or zero-exclusion bounds.
- **Data-quality quirks.** Some slices carry missing channels (`#N/A`), an occasional numeric column header, and macro level-series whose raw magnitudes can distort contribution accounting — all of which the engine must handle defensively.
- **Scale & performance.** Fitting and diagnosing potentially hundreds of Brand × Channel models, then serving them interactively, requires the pipeline to be both fast and memory-disciplined.
- **Definitional dependencies.** Some design choices (the precise "due-to" definition, preferred adstock decay rates, treatment of macro levels, must-include drivers) depend on sponsor guidance and are tracked as open questions.
- **Fixed academic timeline.** The project must reach a presentable state by the final presentation on **August 25, 2026**, leaving limited slack.

1.4 What is needed to complete the project

- The anonymized client workbook and data dictionary (received) plus ongoing sponsor access for check-ins (~weekly, starting July 1, 2026).
- A Python data-science stack (pandas, NumPy, SciPy, statsmodels) and a dashboarding framework (Plotly / Dash or Streamlit) for the UI.
- A three-person team across roughly ten weeks, organized into the phases below, plus faculty advisement and sponsor review.
- No specialized hardware: the work runs on standard laptops; no paid data or external equipment is required.

1.5 Benefits to the organization

- **Speed & scale.** Model an entire Brand × Channel portfolio from one pipeline rather than slice-by-slice, compressing weeks of manual modeling into a same-day run.
- **Consistency & auditability.** Every model uses the same leakage rules, adstock treatment, selection logic, and diagnostics — making results reproducible and defensible to the client.
- **Business credibility.** Sign-constrained final fits guarantee coefficients tell a sensible story (media and distribution lift volume; price and competitive pressure suppress it), which is what wins client trust.
- **Self-serve review.** The UI lets non-technical analysts rerun, tweak, and finalize models, freeing data scientists for higher-value work.
- **Reusable asset.** Big Chalk gains a modular engine it can apply to future CPG engagements beyond this single client.

2 Project Proposal

2.1 General information & background

Big Chalk Analytics sponsors this Northwestern MSDS capstone to design and build an automated marketing-mix regression solution for a national CPG client. The client sells many niche/flavored products across multiple retailer channels; the business question is *what drives weekly Volume Sales* for each Brand $\times$ Channel slice, and by how much. The project began with a kickoff in June 2026, an anonymized three-year weekly dataset, and a clear brief from the sponsor's data-science team. It runs from **June 22 through the end of August 2026**, with the final presentation on **August 25, 2026**.

2.2 Challenges to be faced

- **Target leakage.** The columns most correlated with Volume Sales (Dollar Sales and the per-merch-condition Volume Sales breakdowns) are algebraic decompositions of the target and must be excluded as predictors, or fit becomes meaninglessly inflated.
- **Sign discipline at scale.** Automated selection must be paired with constraints so coefficients respect expected business signs across every slice, not just on a hand-tuned example.
- **Media carryover.** Media spend influences sales beyond the week it is spent, so spend series must be adstocked/decayed before modeling.
- **Collinearity & inference.** Marketing variables are correlated (high VIF) and weekly sales exhibit residual autocorrelation, both of which threaten stable, interpretable coefficients.
- **Heterogeneity.** Hundreds of slices vary in available channels and data quality, so the engine must degrade gracefully rather than fail on edge cases.

2.3 Alternatives considered

- **Single pooled / hierarchical model** — rejected as the primary deliverable: it obscures the slice-level coefficients the client needs, though it remains a useful cross-slice sanity check.
- **Pure Ridge / Lasso final model** — rejected for the *final* fit because it cannot honor sign or zero-exclusion bounds; retained as an *exploration* tool during the build phase.
- **Tree ensembles (random forest / gradient boosting)** — strong for variable importance and exploration, but not directly interpretable as the additive coefficient + contribution output the client expects; used to inform selection, not as the deliverable model.
- **Chosen approach — a two-stage engine:** automated selection (VIF pruning + forward stepwise) to choose variables, followed by **sign/bound-constrained least squares** for the final fit. This satisfies both automation and the sponsor's constraint requirement.

2.4 Project objective (vision & mission)

Vision: give Big Chalk a repeatable, trustworthy way to model sales drivers at full portfolio granularity. **Mission:** deliver an automated regression engine plus an interactive UI that produces sign-sensible, fully diagnosed Volume Sales models for every Brand × Channel combination, reviewable and finalizable by non-technical analysts.

2.5 Project results & activities

Result 1 — A validated, reusable automated regression engine

- **Activity 1.1 — Data understanding & leakage control.** Profile the workbook, codify predictor families and expected signs, and exclude target decompositions (Volume/Dollar Sales breakdowns).
- **Activity 1.2 — Feature engineering.** Adstock media spend (geometric decay), attach sign priors by family, and standardize predictors for selection.
- **Activity 1.3 — Two-stage estimation & diagnostics.** VIF prune → forward stepwise selection → constrained least-squares final fit; emit coefficients, fit metrics, VIF, t-stats, additive contributions, and fit/residual charts.

Result 2 — Portfolio-scale automation & an interactive modeling UI

- **Activity 2.1 — Scale to all slices.** Loop the engine over every Brand × Channel combination and assemble a portfolio results table with comparable diagnostics.
- **Activity 2.2 — Dashboard build.** Data upload → automated model design → results, with drill-down to any single slice, charts, and exportable outputs.
- **Activity 2.3 — Human-in-the-loop controls.** Let users include/exclude variables, set coefficient bounds, adjust adstock decay, rerun, and finalize a model — all without code.

2.6 Implementation modalities

Timeline & milestones

Phase	Focus	Window (2026)	Milestone / deliverable
P1	Understand the data	Wk 1 · Jun 22–28	Data profile, leakage rules, predictor-family map
P2	Automate one model end-to-end	Wk 2 · Jun 29–Jul 5	Working engine on a reference slice; sponsor check-in (Jul 1)
P3	Scale to all Brand × Channel models	Wk 3 · Jul 6–12	Portfolio run + comparable results table
P4	Design & build the UI / dashboard	Wks 4–7 · Jul 13–Aug 9	Interactive dashboard with manual override
P4.5	Integration, hardening & UAT	Wk 8 · Aug 10–16	End-to-end tested, sponsor-validated build
P5	Final presentation & handoff	Wks 9–10 · Aug 17–25	Final presentation — Aug 25, 10 AM

Monitoring & controlling

Progress is tracked through roughly weekly sponsor check-ins (beginning July 1) and faculty advisement. Work is version-controlled in Git (dataset excluded per NDA), with a shared findings document logging methodology decisions, results, and the open-question list for the sponsor. Each phase has an explicit milestone that gates the next.

Sustainability & exit strategy

The engine is delivered as modular, documented Python (loader, adstock, selection, constrained fit, diagnostics, contributions) so the UI calls the same functions rather than re-implementing them. At handoff Big Chalk receives the codebase, the UI, a README/runbook, and the findings document, enabling the team to apply the asset to future CPG engagements without the original authors.

Visibility & outreach (audience)

Primary audience: Big Chalk's data-science team (Alex Hathcock and colleagues) and, through them, the end CPG client's analysts. Secondary audience: Northwestern faculty and capstone reviewers. The dashboard is designed so a non-technical reviewer can operate it directly.

2.7 Risk management

Risk	Response / control measure
Coefficients show wrong business signs	Sign-prior table per family + constrained least-squares final fit; flag and prune dead predictors driven to their zero bound.
Target leakage inflates fit	Hard-exclude all Volume Sales * and Dollar Sales * decomposition columns from the predictor set.
Macro level-series distort contributions	Normalize / index or first-difference raw macro levels; present demeaned contribution views; confirm treatment with sponsor.
Residual autocorrelation biases inference	Time-based holdout validation; consider lagged terms or Newey–West/HAC standard errors in Phase 3.
Edge-case slices (missing channels, bad headers)	Defensive loader; per-slice error capture so one bad slice never halts the portfolio run.
Scope creep on UI vs. fixed timeline	MoSCoW-prioritized requirements; ship Must-haves first, defer Could-haves; phase milestones gate progress.
Open definitional questions block design	Maintain a sponsor question list; default to sensible assumptions and revise after each check-in.

2.8 Other issues

The dataset is anonymized and NDA-protected; no personally identifiable or client-identifying information is exposed, and the data is kept out of any shared repository. The deliverable is an internal analytics tool with no consumer-facing or socio-cultural impact concerns. No external budget is required beyond standard student compute; all libraries used are open-source.

3 Detailed Requirements

Requirements are prioritized with MoSCoW tags: **MUST** **SHOULD** **COULD**. Functional requirements (FR) describe what the system does; non-functional requirements (NFR) describe how well it does it.

3.1 Stakeholders & their needs

Stakeholder	Primary need
Big Chalk data scientists (Alex Hathcock)	A reliable, auditable engine that scales modeling across the portfolio and produces client-ready diagnostics.
End CPG client analysts	Sign-sensible coefficients, contributions ("due-tos"), and fit metrics they can interpret and trust.
Non-technical reviewers	A self-serve UI to rerun, tweak, and finalize models without writing code.
Capstone team & faculty	A modular, documented, reproducible codebase that can be maintained and extended.

3.2 Functional requirements

ID	Priority	Requirement
FR-1	MUST	Ingest the client workbook at Brand × Channel × Week grain, parsing each brand tab and the data dictionary, and handle quirks (missing channels, stray numeric headers) without failing.
FR-2	MUST	Automatically exclude target-leakage columns (all Volume Sales * and Dollar Sales * decompositions) from the predictor set.
FR-3	MUST	Apply geometric adstock/decay to media-spend series, with a configurable decay rate per channel.
FR-4	MUST	Assign expected sign priors by predictor family (price & competitive → negative; distribution, trade, media, seasonality → positive; macro → unconstrained).
FR-5	MUST	Perform automated variable selection: prune by VIF, then forward-stepwise entry at a configurable significance threshold.
FR-6	MUST	Fit a final model with per-coefficient constraints (positive / negative / unconstrained / custom range excluding zero) via constrained least squares.
FR-7	MUST	Prune dead predictors driven to their zero bound and refit, so the final model contains only contributing drivers.
FR-8	MUST	Output, per model: coefficients (with sign & family), fit metrics (R^2 , adjusted R^2 , in-sample & holdout MAPE, Durbin–Watson), VIF, t-stats, and additive contributions ("due-tos") that reconstruct the prediction.
FR-9	MUST	Validate each model on a time-based holdout (train on earlier weeks, score later weeks) rather than random k-fold.
FR-10	MUST	Run the full engine over every Brand × Channel combination and assemble a comparable portfolio results table.
FR-11	MUST	Provide a UI to upload data, trigger automated modeling, and view results with drill-down to any single Brand × Channel model.
FR-12	SHOULD	Let users manually include/exclude variables, set coefficient bounds, and adjust adstock decay, then rerun and finalize a model from the UI.
FR-13	SHOULD	Render actual-vs-fitted, residual, and driver-importance/contribution charts in the UI.
FR-14	SHOULD	Export model outputs (coefficients, metrics, contributions) to CSV/Excel for client reporting.
FR-15	COULD	Offer exploration estimators (Ridge/Lasso, tree importance) side-by-side to inform variable selection.
FR-16	COULD	Support must-include drivers the client always expects (e.g., own price, distribution) even when selection would drop them.
FR-17	COULD	Auto-fit adstock decay rates rather than using fixed defaults.

3.3 Non-functional requirements

ID	Category	Requirement
NFR-1	Performance	A single Brand × Channel model fits in seconds; a full portfolio run completes within a few minutes on a standard laptop.
NFR-2	Scalability	The same functions serve one slice or hundreds without re-implementation; the UI calls the engine API directly.
NFR-3	Reliability	Per-slice error isolation: a single failing slice is logged and skipped, never halting the portfolio run.
NFR-4	Interpretability	Every delivered coefficient carries an expected sign and family; contributions reconstruct the prediction to within negligible numerical error.
NFR-5	Usability	A non-technical reviewer can run and tweak a model end-to-end through the UI without writing code.
NFR-6	Maintainability	Modular, documented Python with a clear engine/UI separation and a README/runbook.
NFR-7	Reproducibility	Fixed random seeds and version-controlled code yield identical results on re-run.
NFR-8	Security & privacy	NDA-protected data is never committed to shared repositories; only derived, anonymized outputs are shared.
NFR-9	Portability	Runs on standard laptops with an open-source stack; no specialized hardware or paid services.

3.4 Constraints & assumptions

- **Constraint:** the final estimator must honor coefficient bounds, excluding pure Ridge from the final fit.
- **Constraint:** data is NDA-protected and anonymized; no redistribution.
- **Assumption:** the supplied weekly grain and ~3-year history are representative of go-forward client data structure.
- **Assumption:** sponsor guidance resolves open definitional questions (exact "due-to" definition, decay rates, macro-level treatment, must-include drivers) during the check-in cadence.

4 Testing Plan

Testing spans four levels — unit, integration, system, and acceptance — plus targeted data-validation and statistical-correctness checks specific to a modeling product. Each test traces back to the requirements in Section 3.

4.1 Testing strategy & environment

Tests are automated with **pytest** and run on every meaningful change. A small anonymized fixture slice and synthetic data with known coefficients serve as ground truth so statistical behavior can be checked deterministically (fixed seeds). The engine and UI are tested separately and then end-to-end. Defects are triaged by severity; a release gate requires all MUST-level functional tests and all statistical-correctness checks to pass.

4.2 Unit testing

Validates individual engine functions in isolation, primarily against synthetic data with known answers.

Component	What is verified	Traces to
Data loader	Correct grain, dtype coercion, and graceful handling of missing channels and stray numeric headers.	FR-1, NFR-3
Leakage filter	All Volume Sales * / Dollar Sales * columns are removed from predictors; legitimate drivers are retained.	FR-2
Adstock transform	Geometric decay matches the closed-form recurrence; decay = 0 reproduces raw spend.	FR-3, FR-17
Sign-prior mapping	Each predictor family maps to the correct expected sign.	FR-4
VIF / stepwise selection	High-VIF predictors are pruned; only predictors below the p-threshold enter on synthetic data with a known structure.	FR-5
Constrained fit	On synthetic data, recovered coefficients respect their bounds and approach known true values; zero-bounded predictors are pruned.	FR-6, FR-7
Contributions ("due-tos")	Additive contributions sum to the prediction within negligible numerical error.	FR-8, NFR-4
Metrics	R ² , adjusted R ² , MAPE, Durbin–Watson match reference implementations.	FR-8

4.3 Integration testing

Validates that components work together along the pipeline and that the UI correctly drives the engine.

- **End-to-end single-slice pipeline:** load → engineer features → select → constrained fit → diagnostics → contributions runs without error and returns a complete model object (FR-1–FR-9).
- **Engine ↔ UI contract:** the dashboard's upload, run, and drill-down actions invoke engine functions with correct arguments and render returned objects faithfully (FR-11–FR-13).
- **Manual-override round-trip:** changing variable inclusion, a coefficient bound, or a decay rate in the UI triggers a correct refit and updated outputs (FR-12).
- **Export integrity:** exported CSV/Excel matches the in-app coefficients, metrics, and contributions (FR-14).

4.4 System testing

Validates the full product against functional and non-functional requirements at portfolio scale.

Test	Pass criterion	Traces to
Full-portfolio run	Engine runs over every Brand × Channel slice, isolating failures, and emits a comparable results table.	FR-10, NFR-3
Performance	Single model fits in seconds; portfolio run completes within a few minutes on a standard laptop.	NFR-1
Sign discipline	Across all slices, delivered coefficients respect their family sign priors (zero sign conflicts).	FR-4, FR-6, NFR-4
Validation discipline	Every model reports a time-based holdout score; no random k-fold leakage of future weeks.	FR-9
Reproducibility	Re-running with fixed seeds reproduces identical coefficients and metrics.	NFR-7
Robustness	Known edge-case slices (missing channels, bad headers) are handled without crashing the run.	FR-1, NFR-3

4.5 User acceptance testing (UAT)

Conducted with the sponsor (Alex Hathcock) against real review tasks. Acceptance is confirmed when the sponsor signs off that:

- A non-technical reviewer can upload data, run automated models, and drill into any Brand × Channel slice unaided (FR-11, NFR-5).
- Coefficients, contributions, and fit metrics are presented clearly and match the client's expected "due-to" semantics (FR-8, FR-13).
- Manual override (variable selection, coefficient bounds, decay) behaves as expected and produces credible refits (FR-12).
- Outputs export cleanly for client reporting (FR-14).

Acceptance criteria summary: all MUST functional tests pass, statistical-correctness checks (contribution reconstruction, sign discipline) pass, performance targets are met, and the sponsor signs off on a self-serve UAT walkthrough.

4.6 Defect management & regression

Defects are logged with severity (blocker / major / minor), reproduced against a fixture, fixed, and covered by a new automated test so they cannot recur. The full pytest suite runs before each milestone demo and before final handoff, providing regression protection as the UI and engine evolve through Phases 3–5.

5 Tools, Technologies & Dataset

5.1 Tools & technologies (with justification)

Tool / technology	Role	Why this choice
Python 3.10+	Core language	Standard for data science; rich modeling ecosystem; team fluency.
pandas / NumPy	Data wrangling	Efficient handling of the multi-tab weekly workbook and matrix assembly.
SciPy (<code>optimize.lsqr_linear</code>)	Constrained final fit	Bounded least squares directly enforces per-coefficient sign/range constraints — the requirement Ridge cannot meet.
statsmodels	Inference & diagnostics	OLS t-stats, VIF, Durbin–Watson, and standard fit metrics for auditability.
scikit-learn	Exploration models	Ridge/Lasso and tree importances as build-phase exploration aids (not the final estimator).
matplotlib / seaborn / Plotly	Visualization	EDA and diagnostic charts; Plotly for interactive in-UI figures.
Plotly Dash / Streamlit	Dashboard / UI	Python-native interactive apps — upload, run, drill-down, and manual override without a separate front-end stack.
openpyxl	Excel I/O	Reads the client .xlsx workbook and writes exportable result files.
pytest	Testing	Automated unit/integration/system tests per Section 4.
Git	Version control	Collaboration and reproducibility; dataset excluded via .gitignore per NDA.

5.2 The dataset

The project uses one anonymized client workbook, *Anonymized Data for Project.xlsx* (NDA-protected): a **General Data Dictionary** tab plus one tab per brand (**Brand 1–10**). The grain is **Channel (Geography) × Brand (Product) × Week**, covering **~3 years of weekly data (~156 weeks, Jan 2023 → Dec 2025)**, up to nine channels per brand, with roughly **1,248 rows and ~113 columns** per brand sheet. The target is **Volume Sales**.

Planned analysis. Phase 1 profiles each slice, establishes the leakage exclusions, and maps predictors to families with expected signs. Subsequent phases engineer features (media adstock, standardization), run the two-stage selection-then-constrained-fit engine per slice, validate on a time-based holdout, and assemble a portfolio-level results table — surfacing coefficients, contributions, and diagnostics through the UI. Documented data cleaning (defensive parsing of missing channels and stray headers) and transformation (adstock, macro-level normalization) accompany every model so the analysis is fully auditable.

6 Data Set Analysis (Phase 1–2 results)

Phases 1 and 2 are already complete. Beyond profiling the dataset, the team has built and validated a first end-to-end automated model, which both proves the methodology and surfaces the data issues that shape the engine's design. This section documents that analysis: profiling and cleaning, the critical leakage finding, feature engineering, the reference-model results, and honest caveats.

6.1 Profiling & data cleaning

Each brand tab was profiled for grain, coverage, dtypes, and completeness. The reference slice (**Brand 1 × Channel 1**) has **156 contiguous weekly observations and no missing values**. Across the workbook the loader handles three real quirks defensively so the portfolio run never halts on a bad slice: **missing/#N/A channels** (e.g., Brand 6), an occasional **stray numeric column header**, and **macro level-series** whose raw magnitudes (tens of millions) can distort contribution accounting.

Key finding — target leakage. The columns most correlated with Volume Sales are *not* usable predictors: **Dollar Sales** (= Price × Volume) and the per-merch-condition **Volume Sales** breakdowns are algebraic *decompositions of the target itself*. Including them would inflate fit while teaching the model nothing causal. The engine **hard-excludes all Volume Sales * and Dollar Sales * columns** — the single most important modeling decision in this phase.

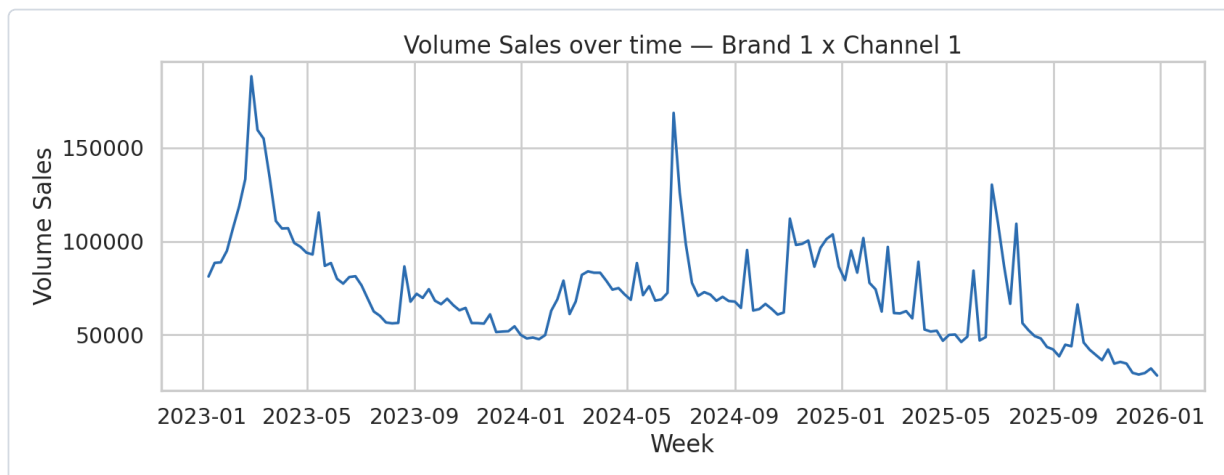


Figure 1 — Weekly Volume Sales for Brand 1 × Channel 1 over 156 weeks (2023–2025), showing trend and recurring seasonal structure.

6.2 Feature engineering & transformation

Three transformations are applied before modeling, each documented and reproducible: **media adstock** — every media-spend series is geometrically decayed ($a_t = x_t + \lambda \cdot a_{t-1}$, $\lambda=0.5$ default) to capture advertising carryover; **sign priors by family** — price and competitive → negative, distribution/trade/media/

seasonality → positive, macro → unconstrained; and **standardization** of predictors for the selection stage. The correlation scan below also motivates the VIF-pruning step, since several marketing variables move together.

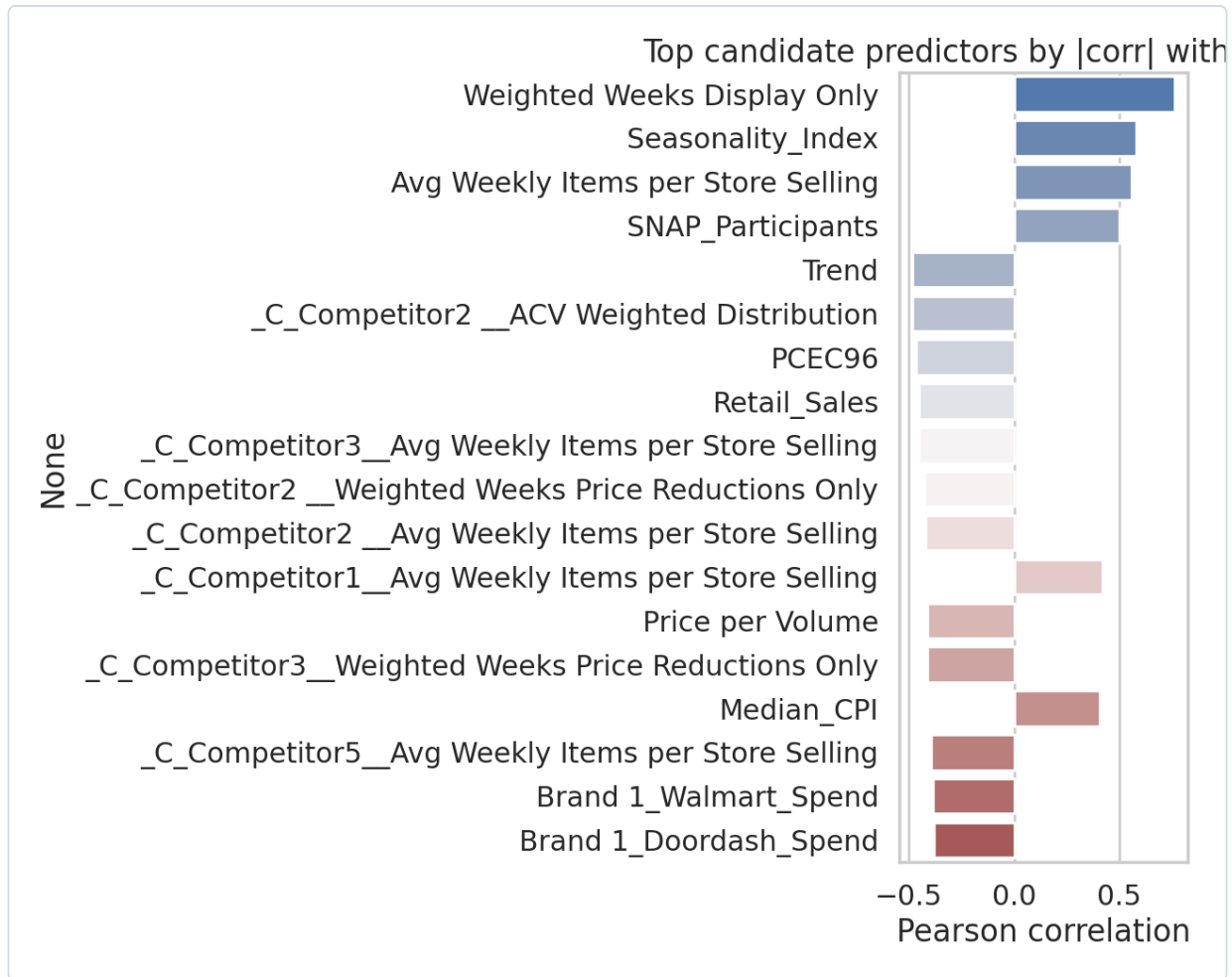


Figure 2 — Predictor correlations with Volume Sales (leakage columns excluded), used to inform family priors and collinearity control.

6.3 Reference model results — Brand 1 × Channel 1

The two-stage engine (VIF prune → forward stepwise → sign-constrained least squares) selected **10 predictors**, every one respecting its expected business sign, all with VIF < 2. Against a naïve train-mean baseline (holdout MAPE ≈ 92%), the model adds large, real predictive value.

Metric	Value	Metric	Value
Predictors selected	10	In-sample MAPE	11.7%
R ²	0.838	Holdout MAPE (26 wks)	19.6%
Adjusted R ²	0.827	Durbin–Watson	0.81
Sign conflicts	0	Max VIF	< 2

Drivers (all signs as expected): trade execution (Weighted Weeks Display, Weighted Weeks Feature) and Seasonality lift volume; competitor display and competitor price reductions suppress it; Meta and Google ad spend contribute positively. Validation uses a **26-week time-based holdout** (train on the first 130 weeks, score the last 26) — appropriate for time series, unlike random k-fold.

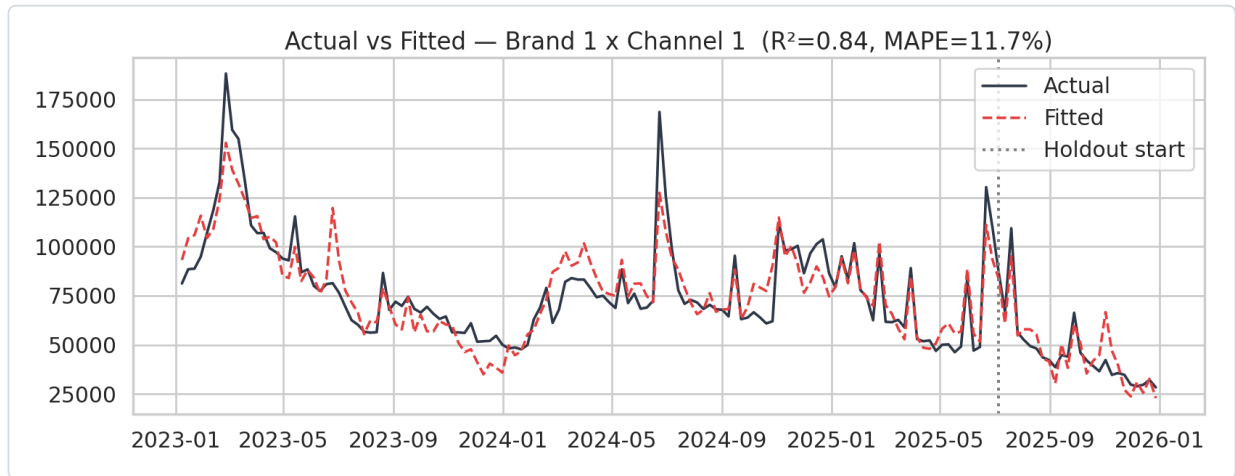


Figure 3 — Actual vs. fitted weekly Volume Sales, in-sample and 26-week holdout.

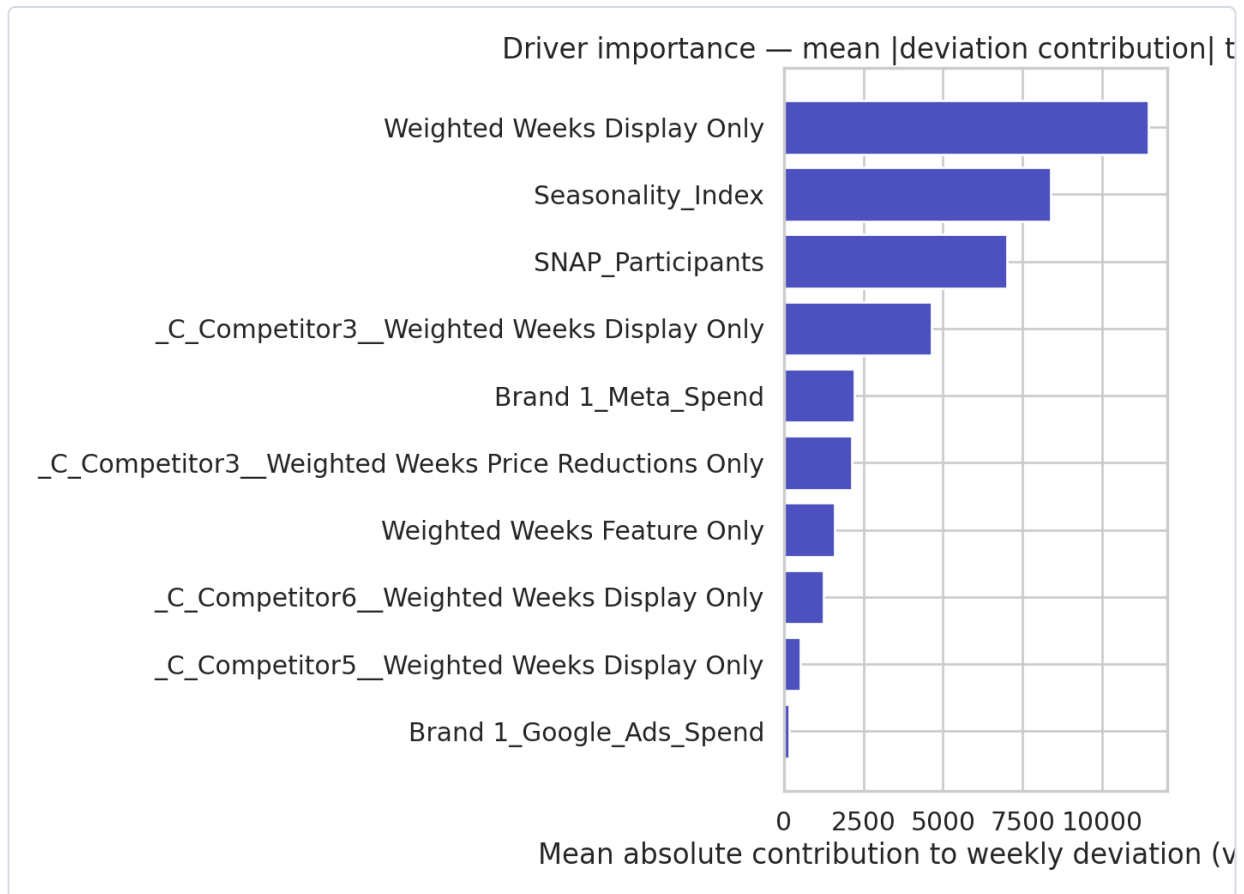


Figure 4 — Additive driver contributions (“due-tos”); contributions reconstruct the prediction to within $\approx 1e-10$.

6.4 Generalization across slices

The identical pipeline runs unchanged on other slices with strong fit — **Brand 2 × Ch1 $R^2=0.90$, Brand 3 × Ch1 $R^2=0.92$, Brand 4 × Ch1 $R^2=0.85$** — giving confidence it will scale to all Brand × Channel combinations in Phase 3 without per-slice hand-tuning.

6.5 Honest caveats & next steps

- **Macro level-series absorbing the baseline.** SNAP_Participants enters with a tiny coefficient but, because its raw magnitude is huge, its contribution effectively sets the model's base level alongside a large negative intercept. Fit is unaffected, but the due-to story is distorted; remedies are to index/normalize or first-difference raw macro levels (charts already show a demeaned contribution view).
- **Residual autocorrelation** (Durbin–Watson ≈ 0.81) is expected for weekly sales but biases standard errors; Phase 3 will consider lagged dependent terms or Newey–West/HAC standard errors.
- **In-sample vs. holdout gap** (11.7% → 19.6%) suggests mild over-fit or a 2025 level shift, to be checked as more slices are modeled.

Conclusion. The dataset supports accurate, sign-sensible Volume Sales models once leakage is removed and media is adstocked. The validated reference model and clean generalization de-risk the Phase 3 portfolio expansion and the Phase 4 UI, which call the same documented functions.

Appendix A Data Dictionary — Predictor Families

Family	Expected sign	Representative variables
Target	—	Volume Sales (dependent variable)
Own price	Negative	Price per Volume (overall & by trade condition)
Category / competitive price	Negative	Category Price per Volume, Total Category Price per Volume
Own distribution	Positive	Avg Weekly Items per Store Selling, ACV Weighted Distribution, Total Points of Distribution
Trade execution	Positive	Weighted Weeks by merch condition (Feature, Display, Feature & Display, Price Reductions, ...)
Media spend (adstocked)	Positive	Meta, TikTok, Google Ads, Walmart, Amazon, Instacart (+ Display/CTV), Doordash, Absco, Ahold, Kroger
Macro-economic (~18)	Unconstrained	Unemployment Rate, CPI, CFNAI, Gas Price, UMCSSENT, T5YIE, PSAVERT, PCEC96, SNAP_*, Fed Funds, Unemp. Claims, Retail Sales
Seasonality & trend	Positive	Trend, Seasonality Index
Competitive	Negative	Competitor 1–6 distribution (Items/Store, ACV) & trade (Weighted Weeks)
Excluded (leakage)	—	All Volume Sales * and Dollar Sales * decompositions of the target